

AI-Powered Network Attack Detection System

Complete Beginner's Guide: What It Is, How It Works, UI Controls & Data Flow

WELCOME TO THE PROJECT GUIDE!

This guide explains the *AI-Powered Network Attack Detection System* in simple, non-technical terms. It is designed for students, examiners, and security analysts to understand **what** the system does, **why** it is needed, **how** each button and screen works, and **what happens behind the scenes** when actions are performed.

1. What is this Project & Why is it Used?

Simple Analogy: Imagine a high-security office building with hundreds of people entering every minute. A traditional security guard checks physical ID badges against a list of known criminals. But what if a criminal dresses up in a custom suit or uses a brand-new fake ID? The traditional guard is fooled.

Our project is an **AI-Powered Digital Security Guard** for computer networks. Instead of just looking at known signatures, it monitors the *behavior, timing, traffic volume, and protocol patterns* of all data entering the system. If it spots suspicious behavior (like someone trying 1,000 door handles a second, or impersonating a trusted coworker), it immediately flags an alert, blocks the attack, and shows security analysts exactly what is happening on a real-time dashboard.

What Problem Does it Solve?

The Problem: Conventional Intrusion Detection Systems (IDS) rely on static rules. Hackers can easily bypass old systems by slightly altering their attack code, varying packet timing, or launching low-volume attacks. Furthermore, old systems trigger hundreds of false alarms, overwhelming security staff.

The Solution: This project uses a **Dual-Engine AI Architecture**:

- **Deep Learning Engine (Neural Network):** Learns complex traffic sequences over time to catch high-volume flooding attacks (DoS, DDoS, Port Scans, Brute-Force).
- **Machine Learning Anomaly Engine (Random Forest):** Monitors network protocol rules to catch sneaky, low-volume eavesdropping attacks (ARP Spoofing / Man-in-the-Middle).

Business & Practical Value

- **Prevents Expensive Server Downtime:** Protects online businesses from crashing during heavy cyberattacks.
- **Protects Sensitive Data:** Prevents hackers from intercepting passwords, credit card numbers, or corporate secrets.
- **Standardized Threat Context:** Automatically maps every attack to global standards (MITRE ATT&CK; & Cyber Kill Chain) so analysts can react instantly.

- **Zero Cost (Open Source):** Built entirely using free tools (Python, Scikit-Learn, Flask, ELK Stack), eliminating commercial SIEM software licensing costs.

2. Technical & Business Terms Made Simple

Term	Simple Plain-English Definition	Real-World Analogy
Network Flow	A sequence of digital data packets sent between two computers during a conversation.	A series of letters mailed back and forth between two pen pals.
DoS / DDoS	Denial of Service. Flooding a server with fake traffic until it slows down or crashes.	10,000 fake customers crowding into a small coffee shop so real customers can't buy coffee.
Port Scanning	A hacker probing all computer ports to discover open services and weak points.	A burglar walking down a street checking every house door handle to find unlocked ones.
Brute-Force	Automated bot guessing passwords repeatedly at high speed until it gets in.	Someone trying every possible 4-digit code combination on a bike lock until it opens.
ARP Spoofing / MITM	Man-in-the-Middle attack. Hacker tricks the local network into routing target traffic through the hacker's laptop.	An eavesdropper rerouting a target's physical mail to their mailbox first, reading it, then re-mailing it.
MITRE ATT&CK;	A world-standard cybersecurity matrix cataloging known adversary tactics and techniques.	A dictionary of known crime methods used by police detectives worldwide.
ELK Stack	Elasticsearch, Logstash, Kibana. A big-data search engine for centralizing and visualizing security logs.	A digital library index system that lets librarians find any security log in milliseconds.

3. Complete User Interface (UI) & Button Guide

This section explains every screen element, navigation item, button, metric card, and interactive control in the SOC Analyst Web Dashboard.

UI Component / Button	What It Shows / What Happens When Clicked	Background Action & Technical Result
Sidebar: Navigation Menu	Contains quick links: Dashboard, MITRE ATT&CK;, Simulator, Alerts, Metrics.	Smoothly scrolls the browser window to the selected panel section.
Pause / Resume Stream Button	Clicking toggles between pausing and resuming the live dashboard feed.	When Clicked: Temporarily suspends JavaScript SSE DOM updates. Background Python engine continues processing and saving events silently.
Stat Cards Bar	Displays 4 live counters: Total Packets, Benign Traffic, Intercepted Attacks, Current Threat Index (0-100).	Updates every 1 second via Server-Sent Events (SSE) stream from Flask backend server.
DoS Attack Button (T1499)	Clicking injects a single-source high-volume Denial of Service attack.	Background Action: Triggers <code>`simulator.set_manual_attack('DoS')`</code> . Generator injects 800-4,000 forward packets/sec with high SYN/RST flags. Deep Learning model flags DoS with ~99% confidence. Threat score rises to ~85.
DDoS Attack Button (T1498)	Clicking injects a coordinated multi-source botnet Distributed DoS attack.	Background Action: Injects 5,000-15,000 packets/sec flood from random botnet IPs. Flow model flags DDoS. Threat score jumps to ~98 (Critical). MITRE T1498 card glows red.
Port Scan Button (T1046)	Clicking injects a rapid port probing reconnaissance attack.	Background Action: Injects short duration packet bursts targeted across random destination ports (1-65535). Model identifies PortScan. Threat score set to ~45 (Medium).
Brute-Force Button (T1110)	Clicking injects repeated failed authentication attempts on SSH/FTP/RDP.	Background Action: Injects flow with 25-120 authentication failure attempts on port 22/21/3389. Model flags BruteForce. Maps to OWASP A07 Auth Failure.
ARP Spoof / MITM Button (T1557)	Clicking injects a protocol-level Man-in-the-Middle spoofing attack.	Background Action: Injects unsolicited ARP reply flood (25-120 replies/sec) and MAC binding conflicts. Random Forest Anomaly Detector intercepts protocol anomaly. Threat score set to ~95.
Timeline & Doughnut Charts	Visual graphs showing danger score over time and category breakdown percentages.	Chart.js re-renders line and doughnut data in real time as new SSE events arrive.
MITRE ATT&CK; Matrix Grid	5 interactive cards representing T1046, T1110, T1557, T1499, T1498.	Flashes neon red for 1.8 seconds whenever an attack matching that technique is intercepted.
Inspect Button (Alert Row)	Clicking opens a pop-up modal showing deep technical packet metrics.	When Clicked: Displays raw Elastic Common Schema (ECS) JSON, exact feature vectors, detection engine name, CAPEC ID, and OWASP recommendations.
Model Performance Panel	Shows accuracy, precision, recall, F1, and FPR for both AI models.	Fetches verified evaluation metrics from <code>`/api/metrics`</code> JSON file.

4. How the Complete Process Works (Step-by-Step)

Here is the complete end-to-end data pipeline from raw network packet to visual dashboard alert:

- **Step 1: Network Traffic Capture / Simulation**

Live network packets (or simulated traffic flows) are continuously captured. Each flow contains timing, packet count, byte rate, port numbers, and protocol headers.

- **Step 2: Dual-Path Feature Extraction**

The *Feature Extractor* converts raw traffic into two numerical feature vectors:

- *Flow Vector (12 features)*: Duration, fwd/bwd packet count, bytes/sec, SYN/RST/ACK flags, failed auth count.
- *ARP Vector (5 features)*: ARP request rate, reply rate, reply/req ratio, MAC change rate, conflict count.

- **Step 3: Dual AI Engine Evaluation**

- **Deep Learning Model (MLP/LSTM)**: Evaluates the Flow Vector and outputs probabilities for Normal, DoS, DDoS, PortScan, and BruteForce.
- **Random Forest Anomaly Detector**: Evaluates the ARP Vector and outputs anomaly score for ARP Spoofing / MITM.

- **Step 4: Ensemble Correlation & Conflict Resolution**

The *Ensemble Correlation Engine* fuses both predictions. If the ARP anomaly score > 60%, it prioritizes ARP Spoofing. Otherwise, it uses the Deep Learning prediction. It computes a unified Threat Score (0 - 100).

- **Step 5: Threat Framework Mapping**

The event is tagged with its MITRE ATT&CK; ID (e.g., T1498), Cyber Kill Chain stage (e.g., Actions on Objectives), CAPEC reference (e.g., CAPEC-125), and OWASP Top 10 category.

- **Step 6: ELK Stack Normalization & Export**

The *ELK Exporter* converts the alert into Elastic Common Schema (ECS 8.x) JSON, writes it to ``logs/network_security_events.json``, and forwards it to Logstash / Elasticsearch.

- **Step 7: Real-Time Web Dashboard Push**

The Flask backend streams the event via Server-Sent Events (SSE) to the browser, updating live charts, incrementing counters, flashing the MITRE matrix, and appending the alert to the live table.

5. Real-World Use Case Examples

Scenario A: E-Commerce Store on Black Friday (DDoS Attack)

A competitor hires a botnet to launch a 10,000 packet/sec flood against an e-commerce website during a sale. Our Deep Learning Flow Model detects the massive packet rate surge and high SYN/RST flags within 1 second, flags a **Critical DDoS Attack (T1498)** with a 98.0 Threat Score, and alerts the network team to block the botnet IPs before the store crashes.

Scenario B: Wi-Fi Eavesdropper in a Corporate Office (ARP Spoofing MITM)

A rogue contractor plugs a laptop into an office Wi-Fi network and sends fake ARP replies to trick executive laptops into sending passwords through the contractor's device. Because ARP spoofing uses low traffic volume, traditional volumetric filters miss it. Our **Random Forest Anomaly Detector** detects the sudden spike in ARP reply/request ratio and MAC address conflict rate (100% precision), flagging an **ARP Spoofing / MITM (T1557)** alert immediately.

SUMMARY FOR EXAMINERS & ANALYSTS:

This system provides end-to-end automated protection: from raw network packet ingestion to AI classification, framework correlation, ELK indexing, and real-time SOC Analyst visual dashboard monitoring.